Refraction of Light Through a Prism

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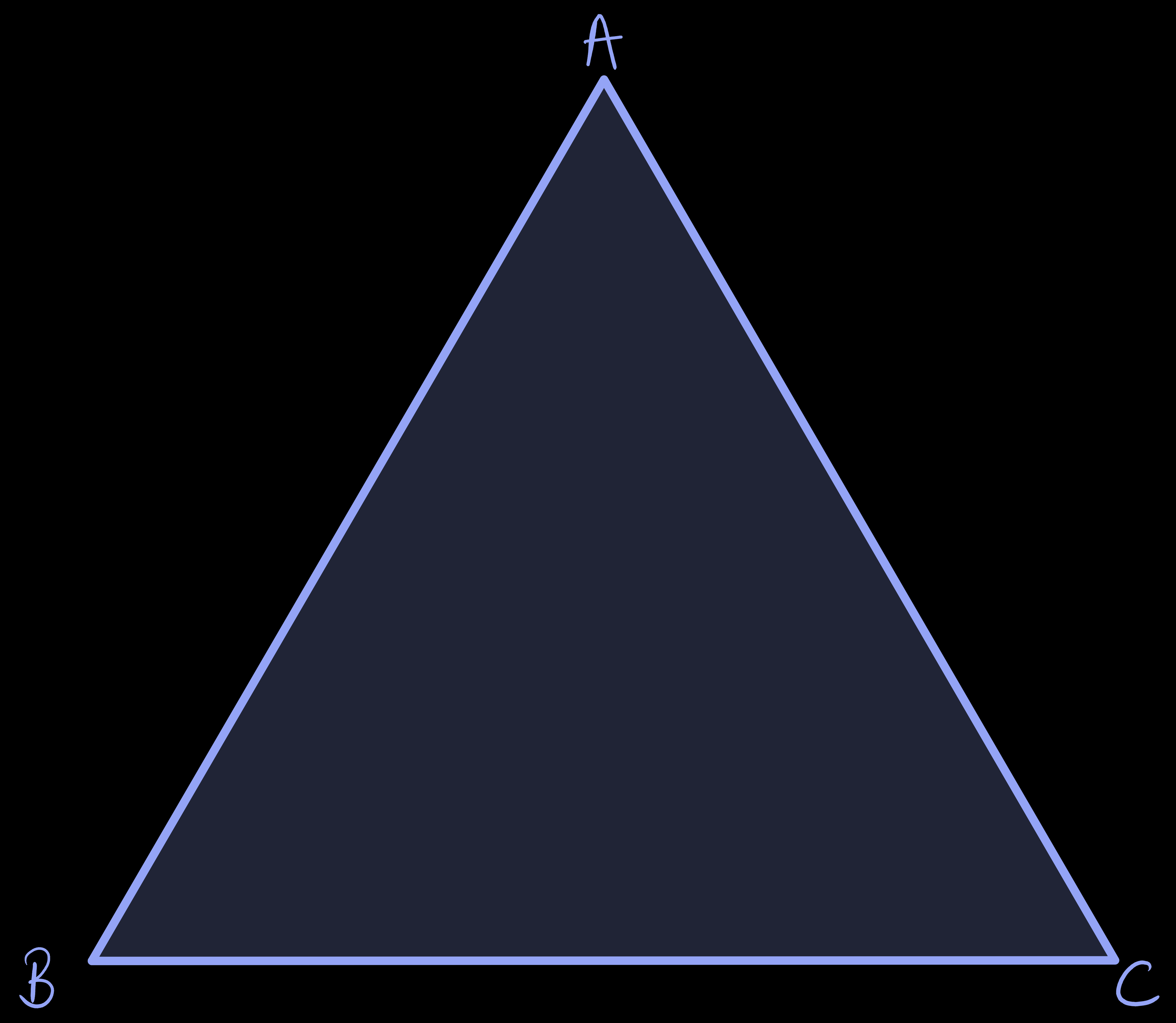
- **Triangular Glass Prism :**

  It has **two triangular bases** and **three rectangular** lateral surfaces. These surfaces are inclined to each other.



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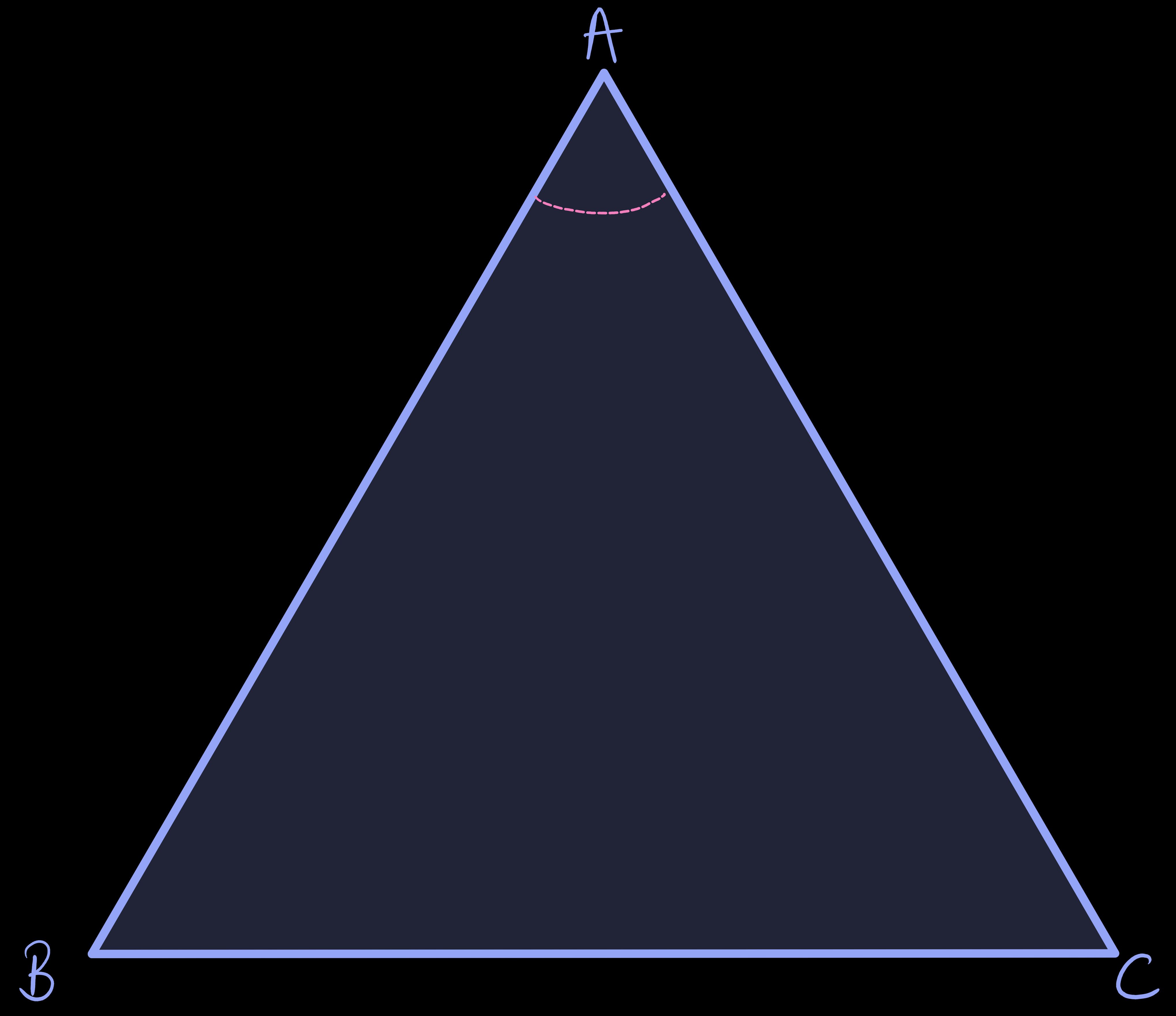
- **Angle of the prism** (A) **:**

  The **angle between its two refracting faces** is called the angle of the prism.



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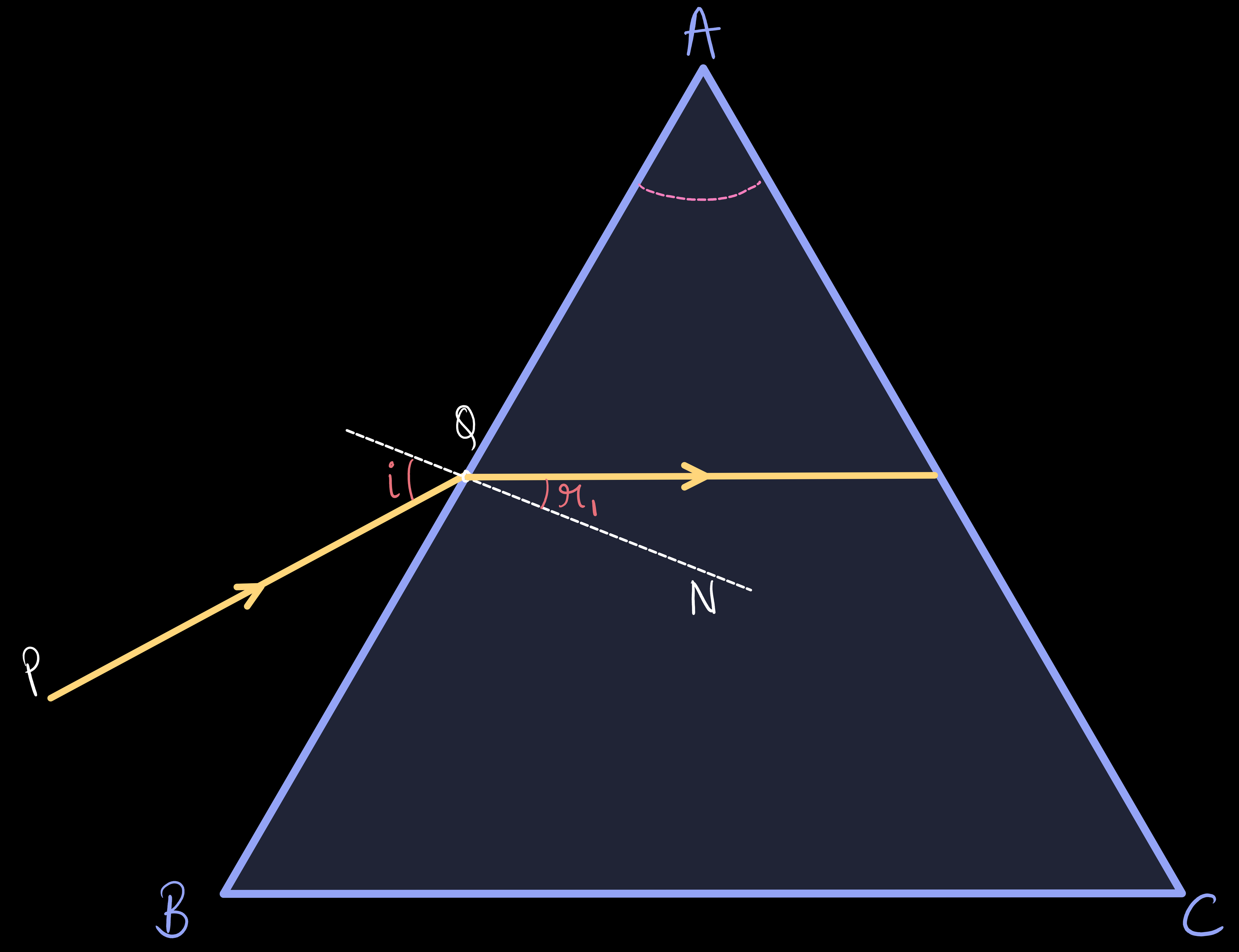
- **Refraction of light by prism :**

  A ray of light is entering **from air to glass** at the **first surface AB**. The light ray on refraction has **bent towards the normal**.



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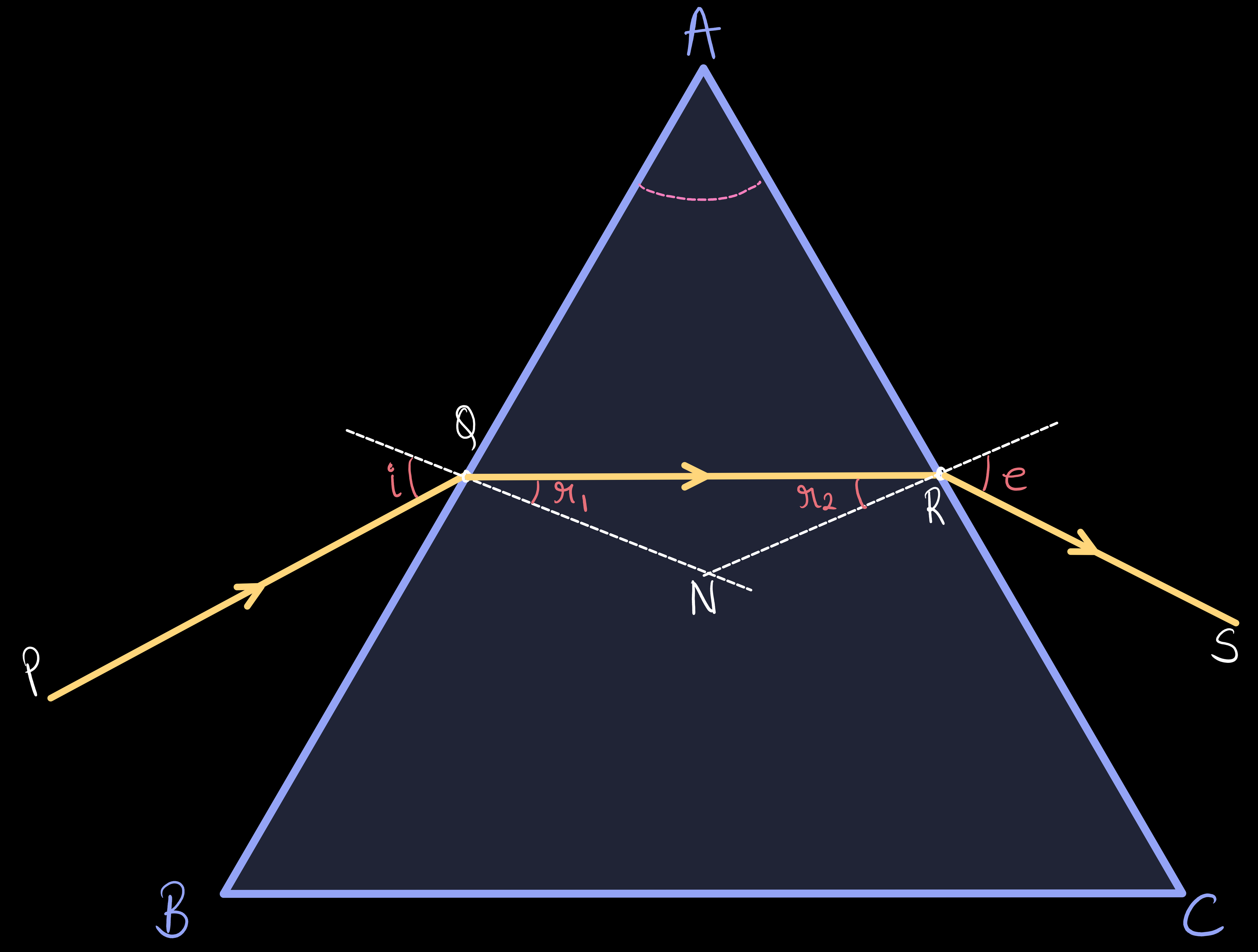
- **Refraction of light by prism :**

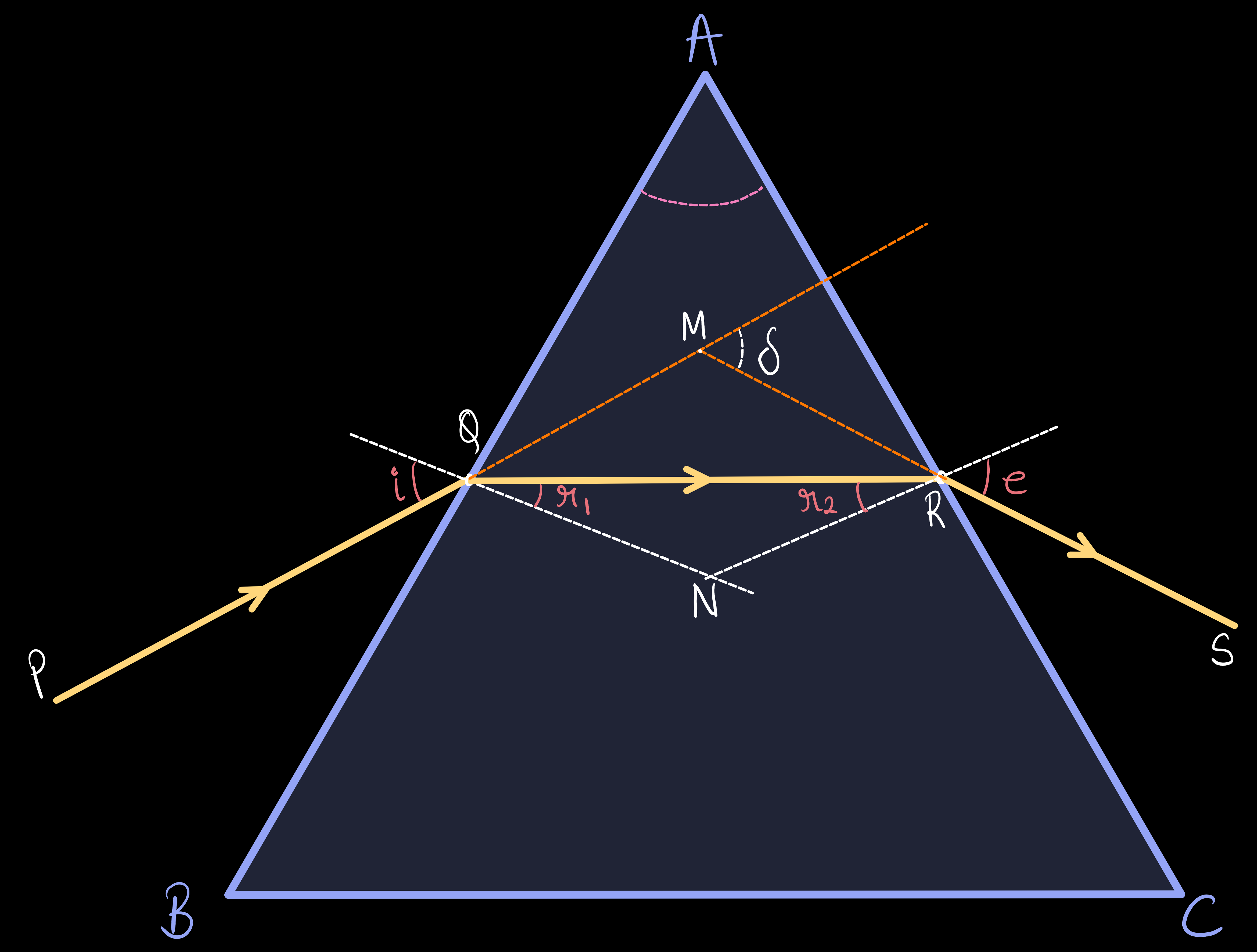
A ray of light is entering **from air to glass** at the **first surface AB**. The light ray on refraction has **bent towards the normal**. At the **second surface AC**, the light ray has **entered from glass to air**. Hence it has **bent away from normal**.



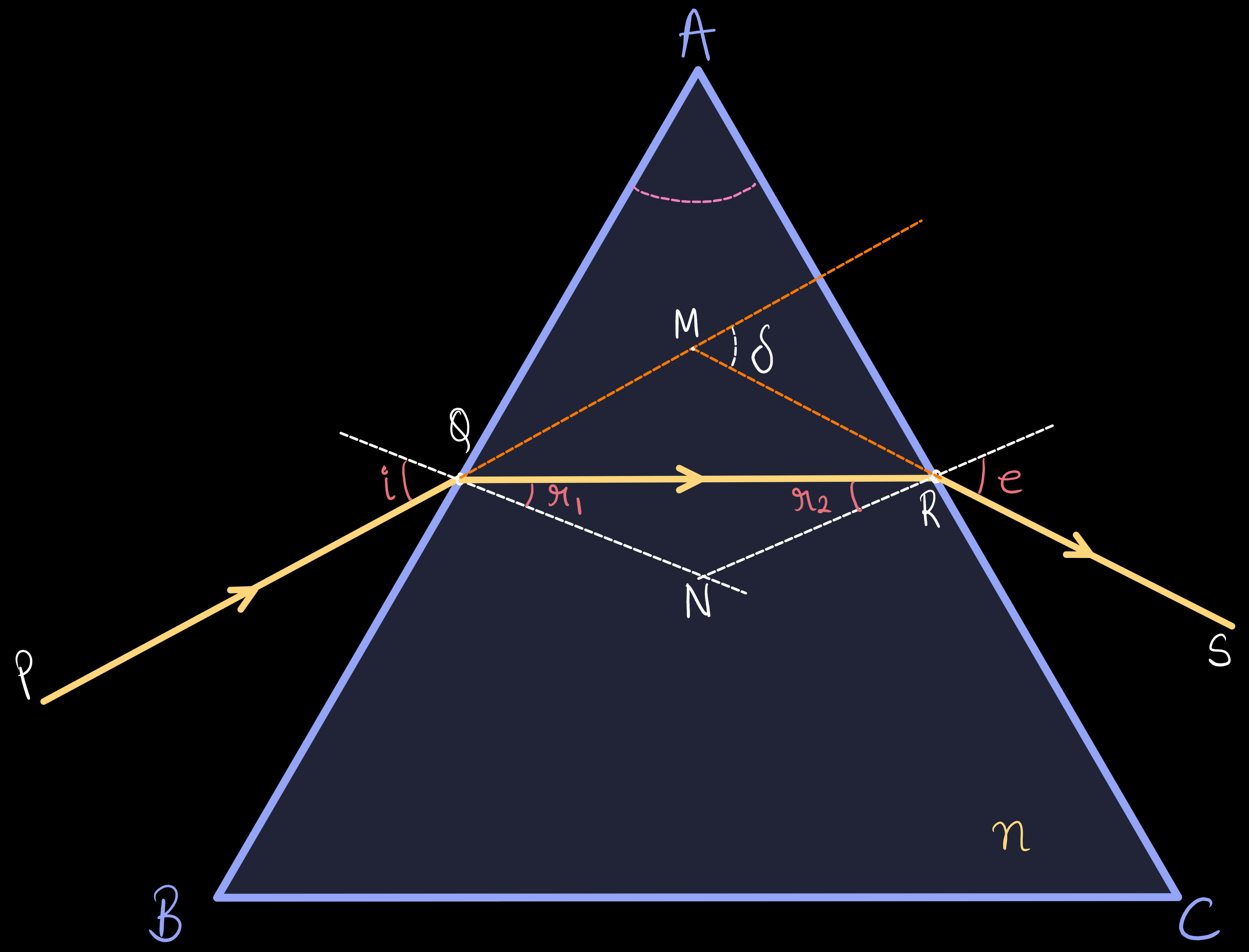
Refraction of Light Through a Prism

- **Angle of Deviation (δ) :**

The **peculiar shape of the prism** makes the emergent ray bend at an angle to the direction of the incident ray. This **angle between the direction of the incident ray and emergent ray** is called the angle of deviation.



Refraction Through a Prism (Derivation)

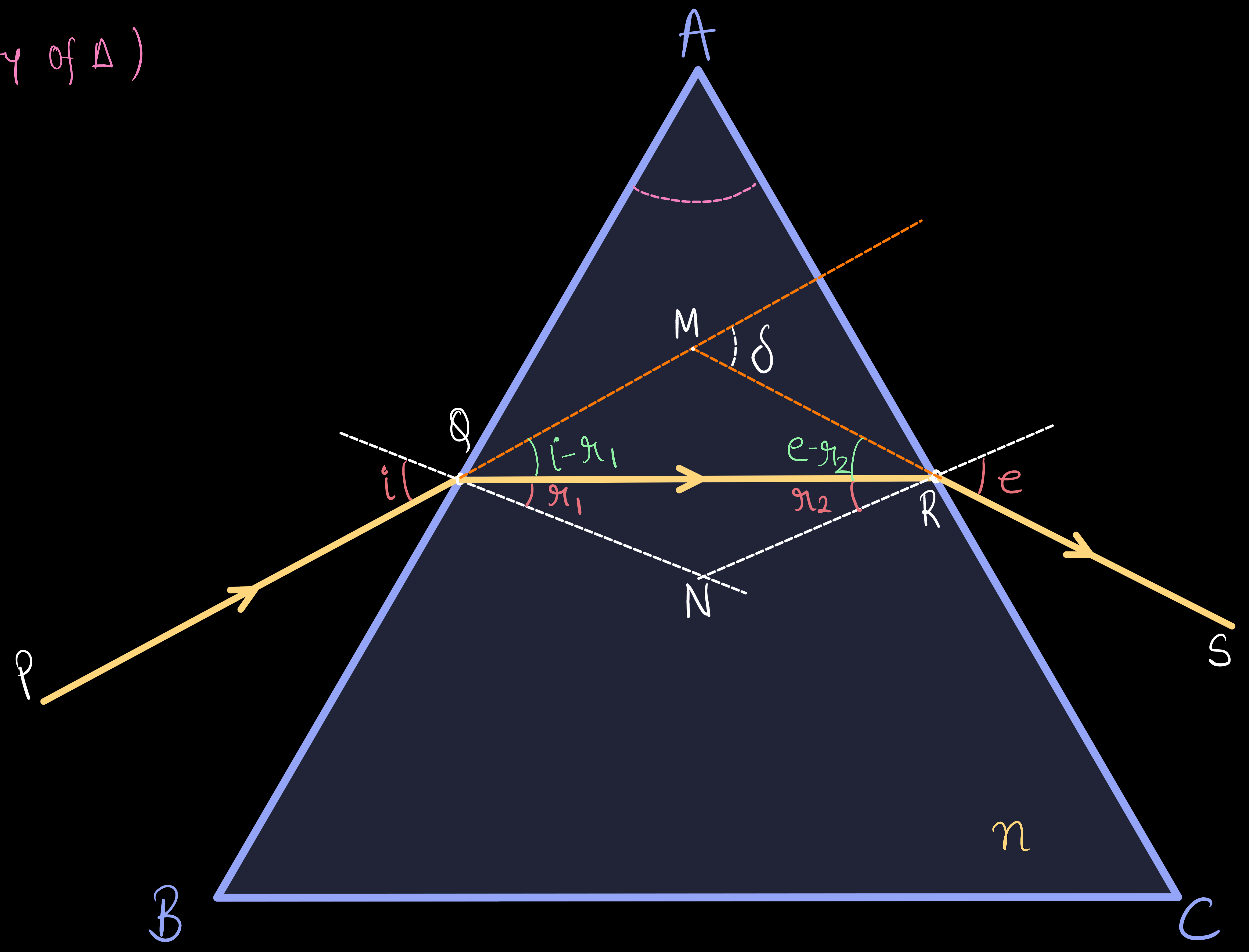


Refraction Through a Prism (Derivation)

In ΔMQR –

$$\delta = i - r_1 + e - r_2 \quad (\text{external Angle property of } \Delta)$$

$$\delta = i + e - (r_1 + r_2) \quad \text{—} \; (1)$$



Refraction Through a Prism (Derivation)

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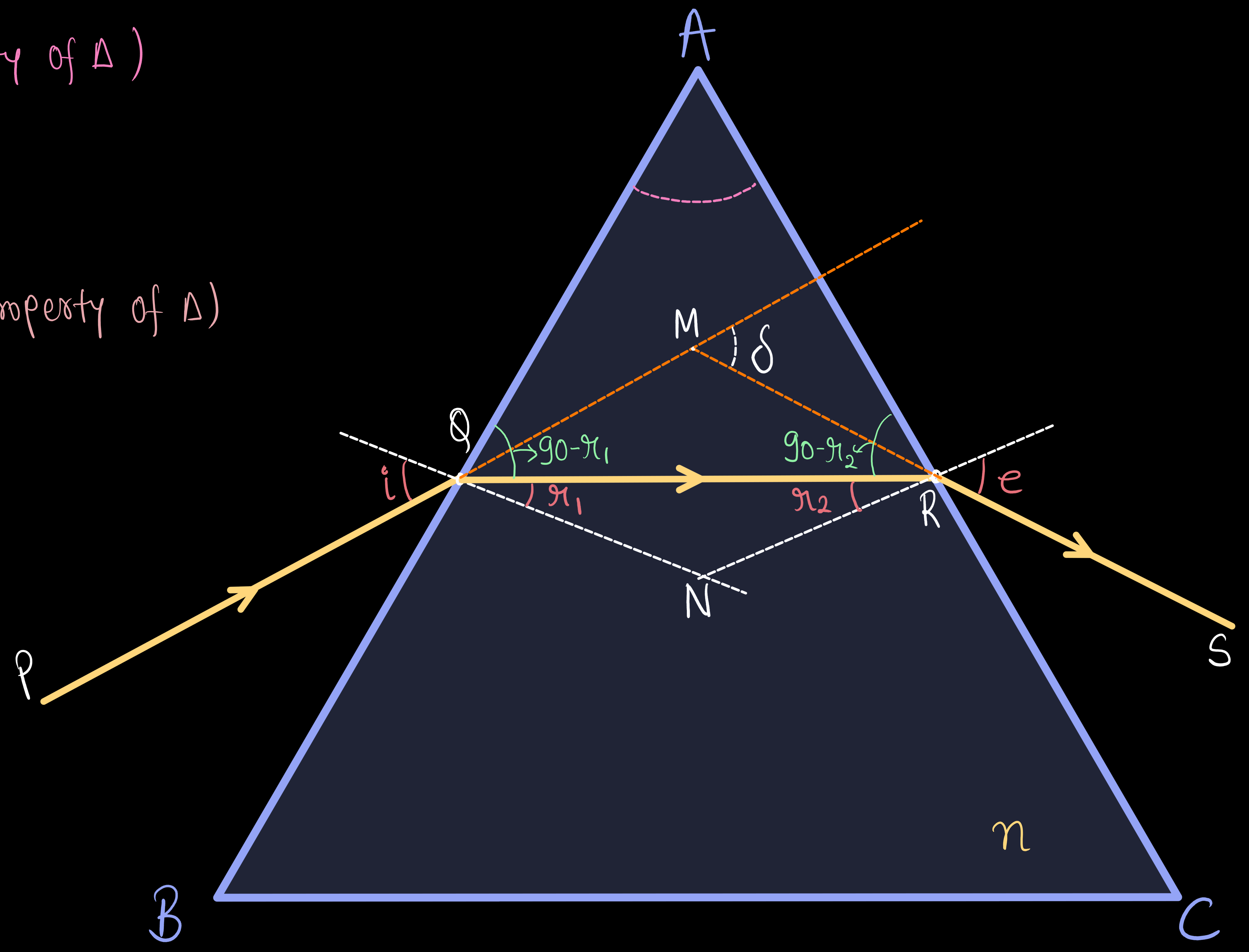
In ΔAQR –

$$A + 90 - r_1 + 90 - r_2 = 180^\circ \quad \text{(Angle sum property of } \Delta)$$

$$A - (r_1 + r_2) = 0$$

$$A = r_1 + r_2 \quad \text{— (2)}$$



Refraction Through a Prism (Derivation)

In ΔMQR –

$\delta = i - r_1 + e - r_2$ (external Angle property of Δ)

$\delta = i + e - (r_1 + r_2)$ — (1)

In ΔAQR –

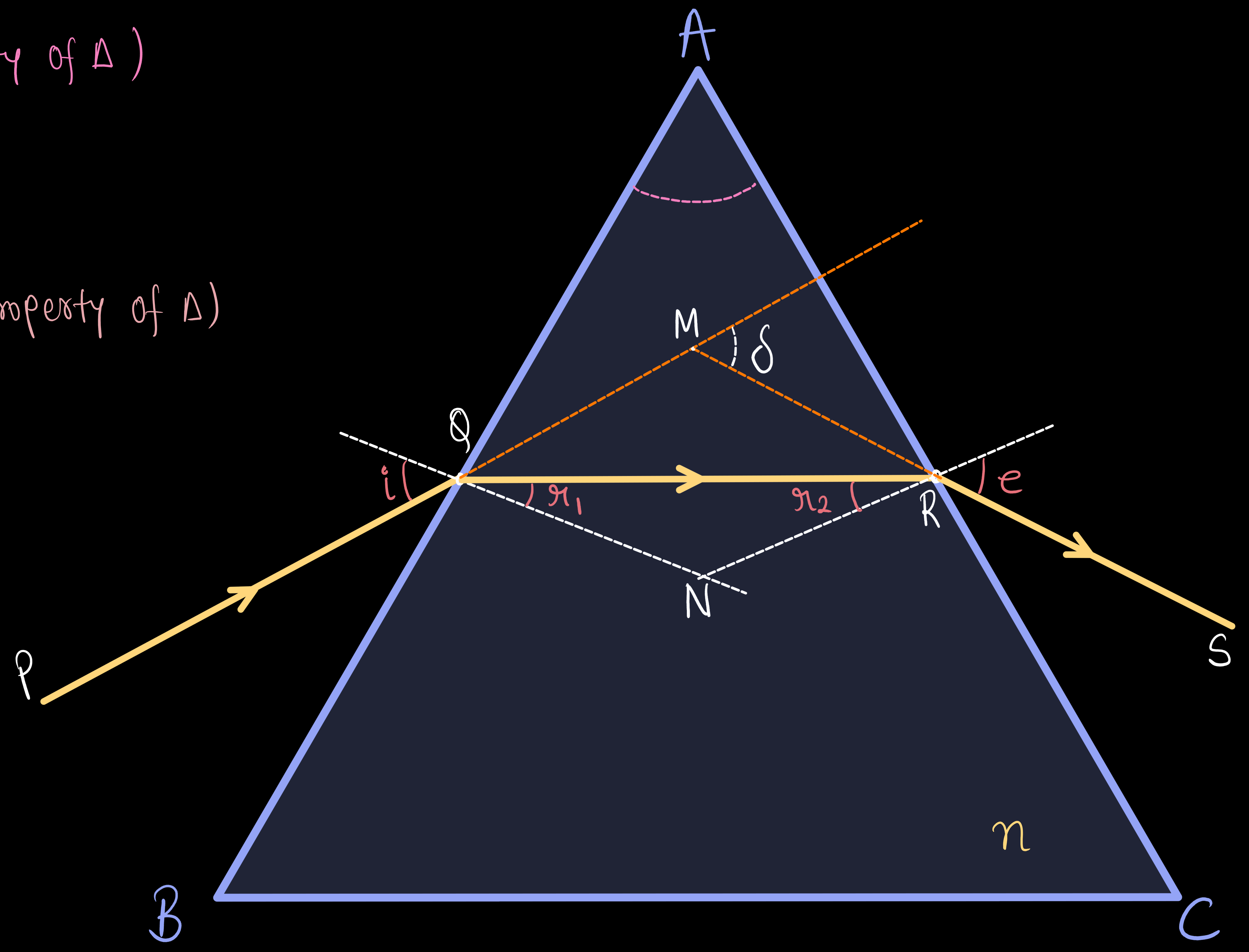
$A + 90 - r_1 + 90 - r_2 = 180^\circ$ (Angle sum property of Δ)

$A - (r_1 + r_2) = 0$

$A = r_1 + r_2$ — (2)

From eq^n (1) & (2)

$\boxed{\delta = i + e - A}$ — (3)



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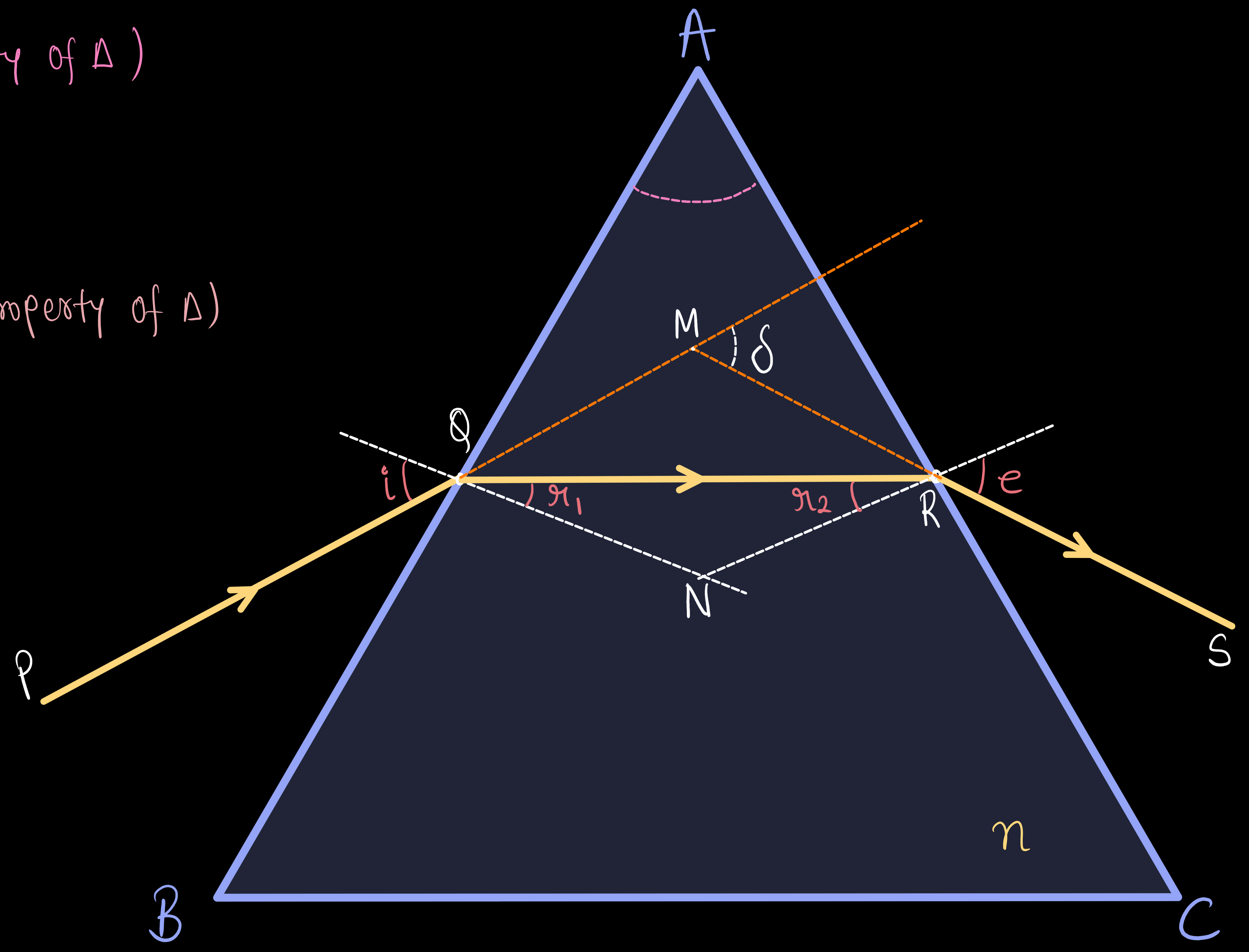
From eq^n (1) & (2)

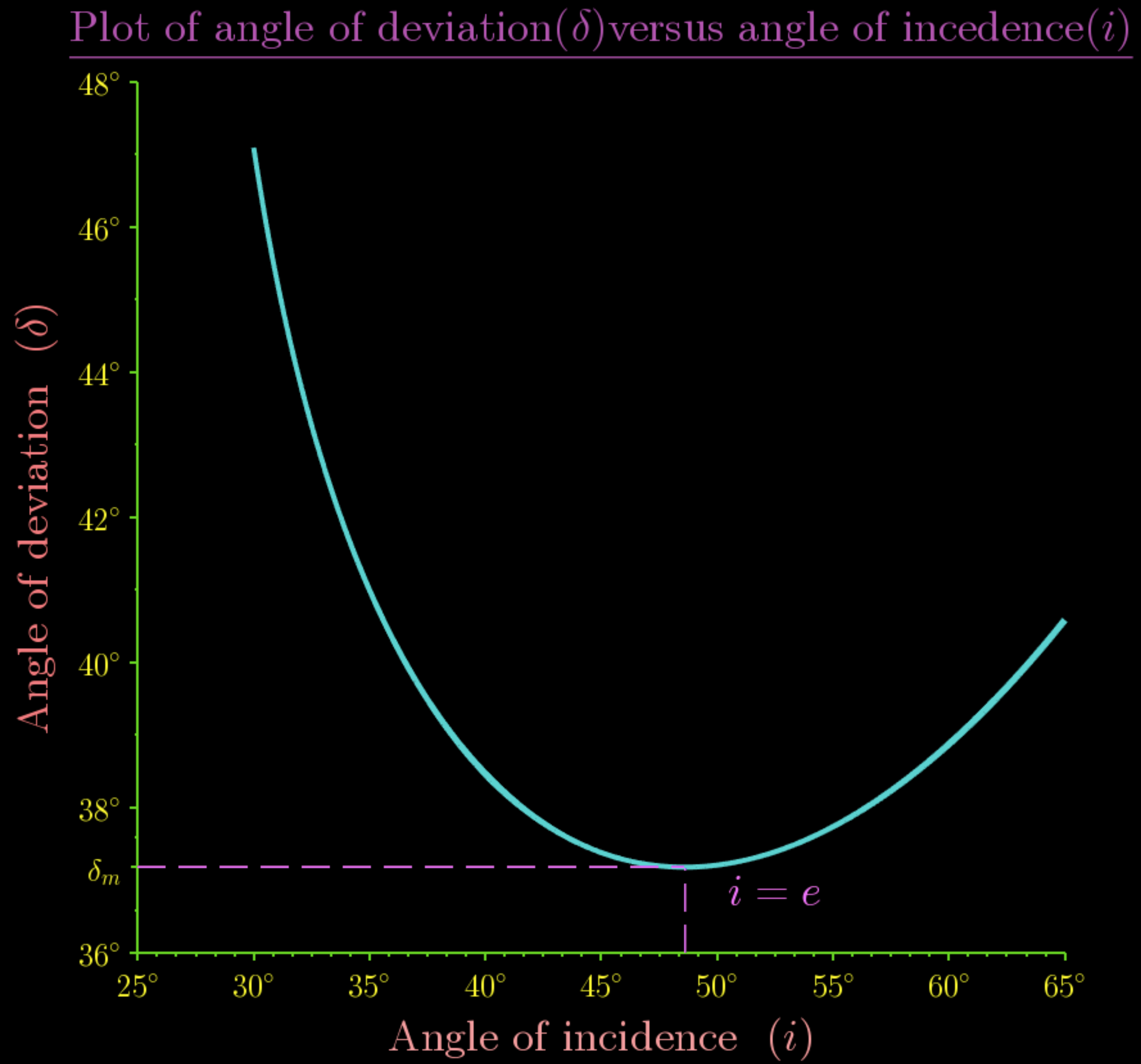
$$\boxed{\delta = i + e - A} \quad \text{— (3)}$$

At minimum deviation position –

$$\delta = \delta_m, \quad r_1 = r_2 = r \quad \text{and} \quad i = e$$

Also, the refracted ray (QR) is parallel to the base of the prism.





Plot of angle of deviation(δ)versus angle of incedence(i)
48°
46°
44°
42°
40°
38°
δ_m
36°
Angle of deviation (δ)
$i = e$
25°
30°
35°
40°
45°
50°
55°
60°
65°
Angle of incidence (i)

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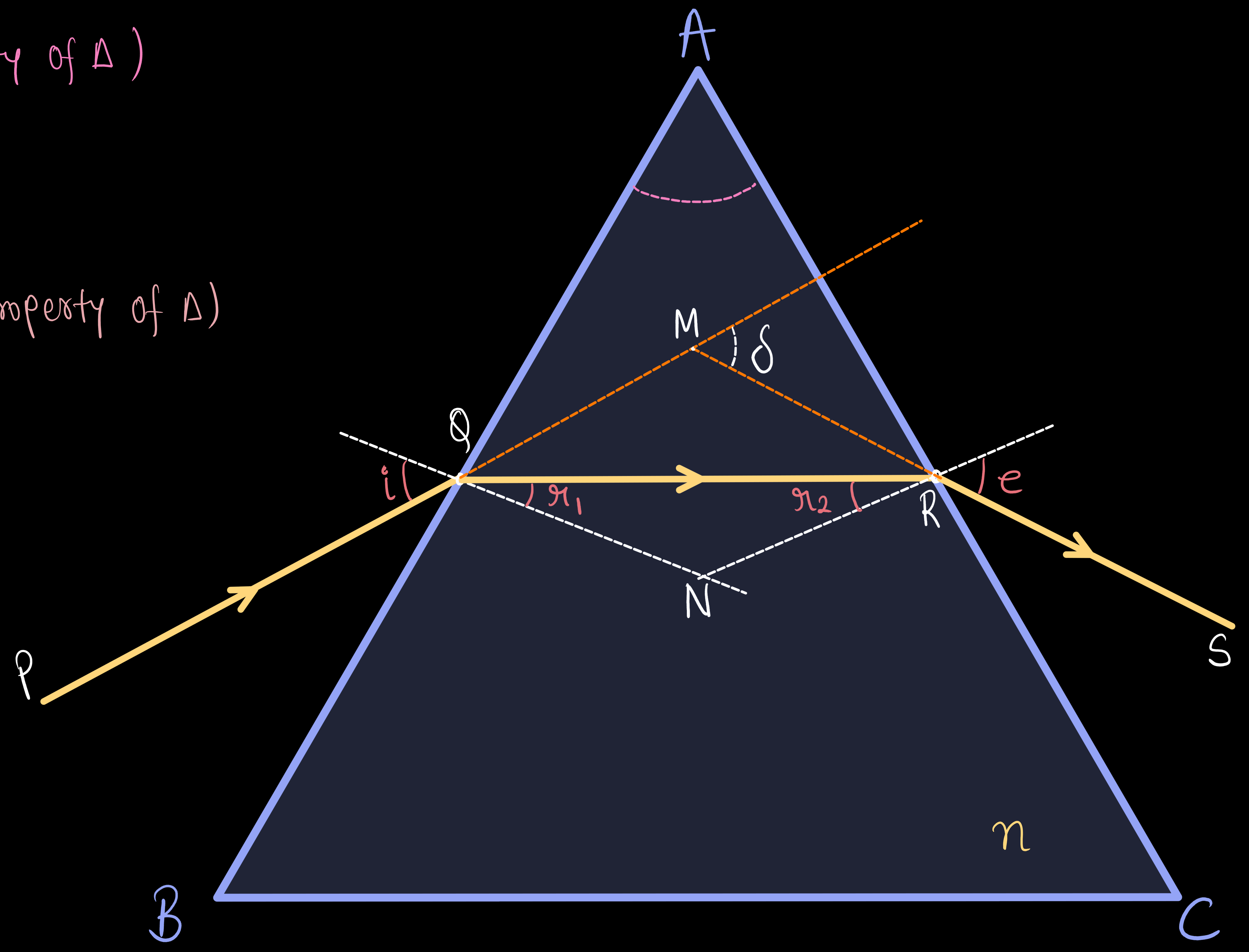
from eq^n (1) $A = r + r$

$A = 2r$

$\boxed{r = \frac{A}{2}}$ — (4)



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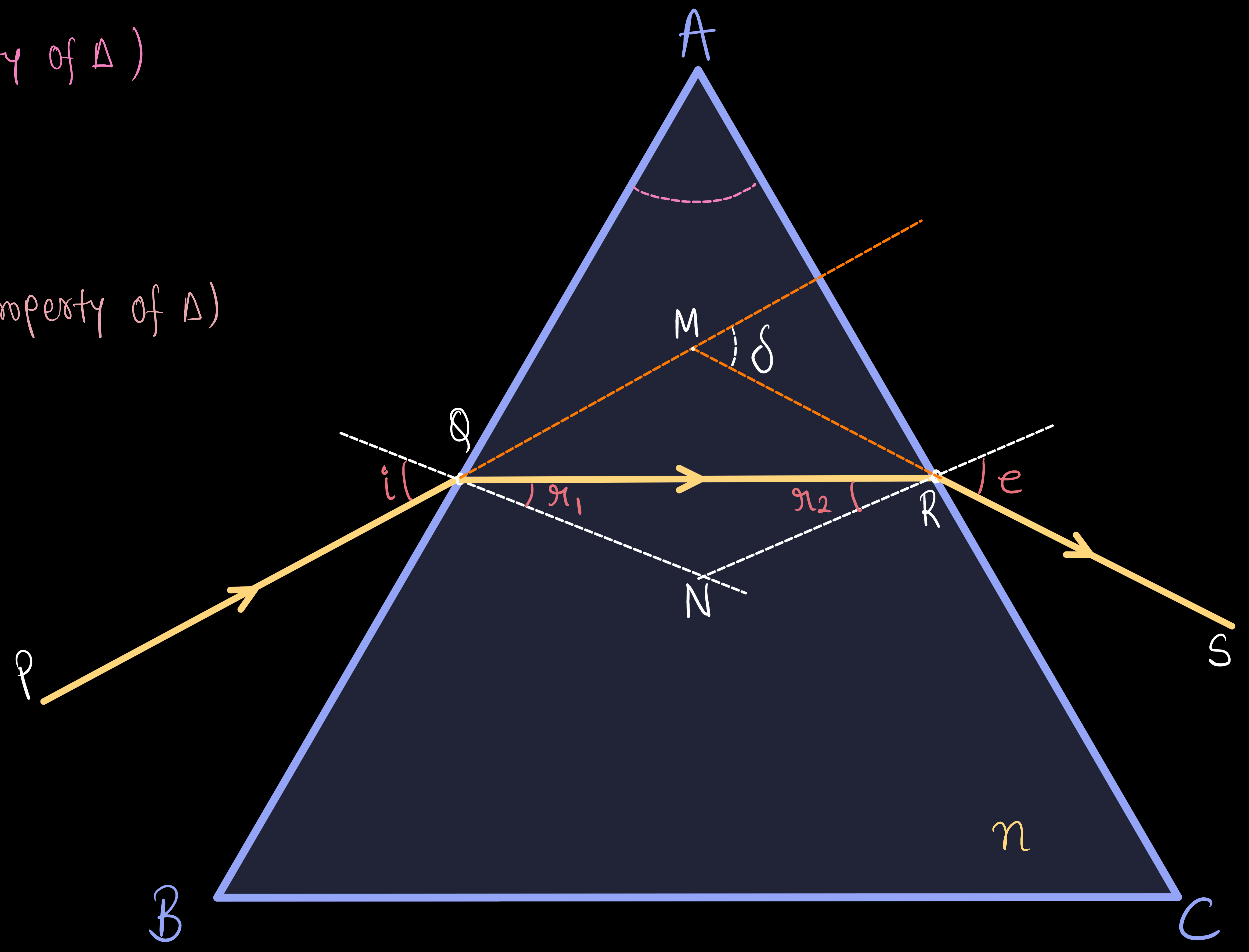

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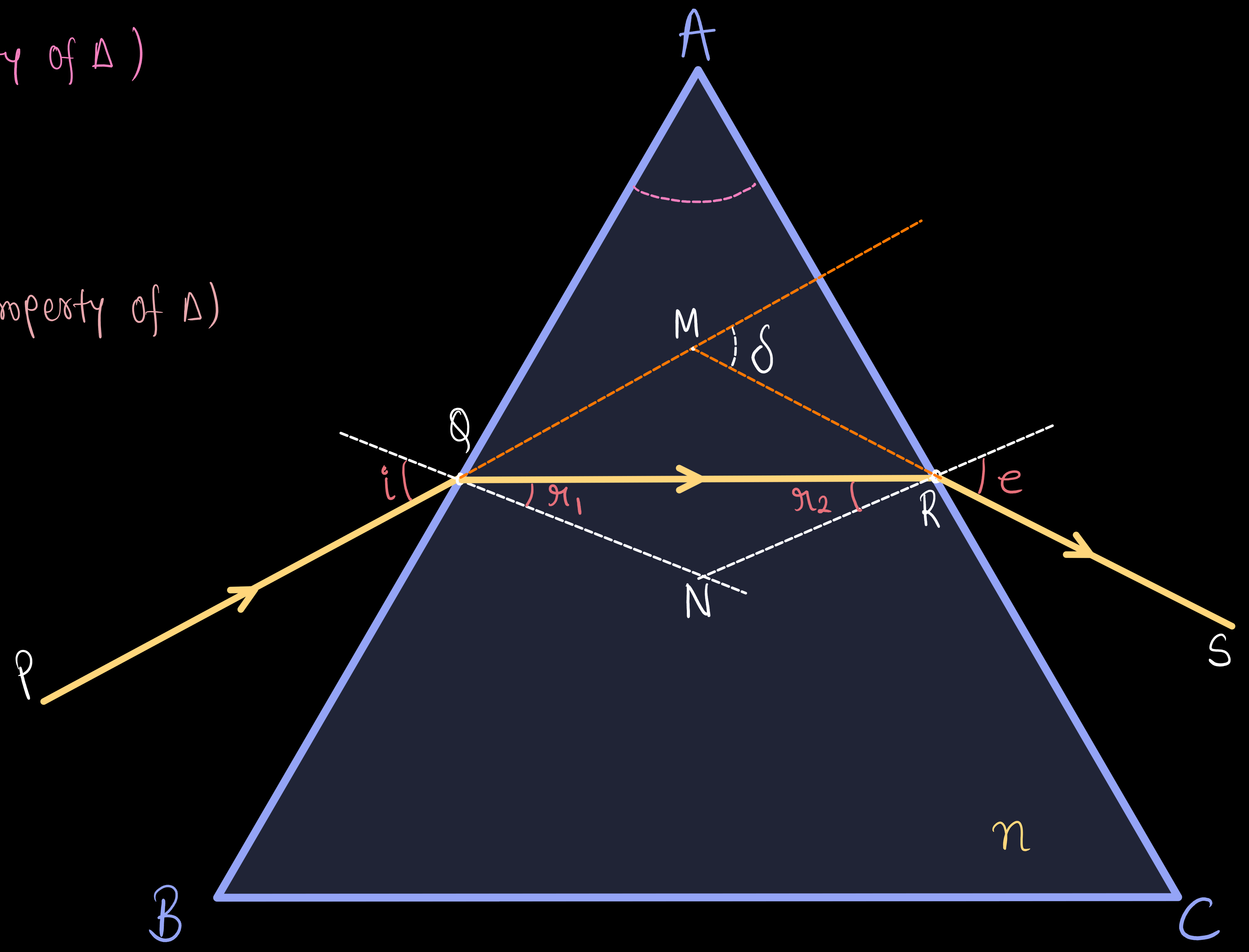

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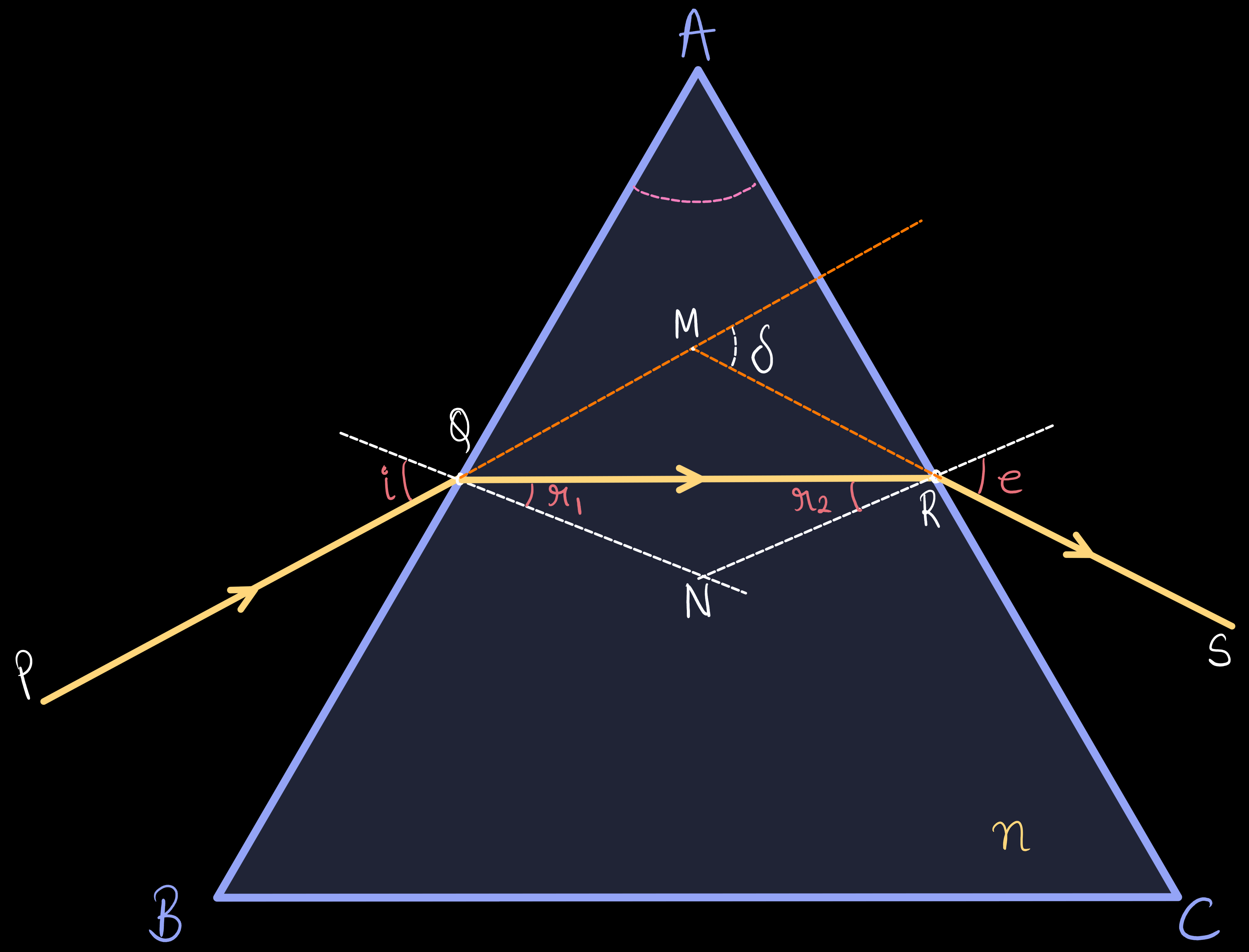
$\delta_m = 2i - A$

$\Rightarrow \boxed{i = \frac{\delta_m + A}{2}}$ — (5)

Refraction Through a Prism (Derivation)

Using Snell's at face AB -

$$1 \times \sin i = n \times \sin r$$



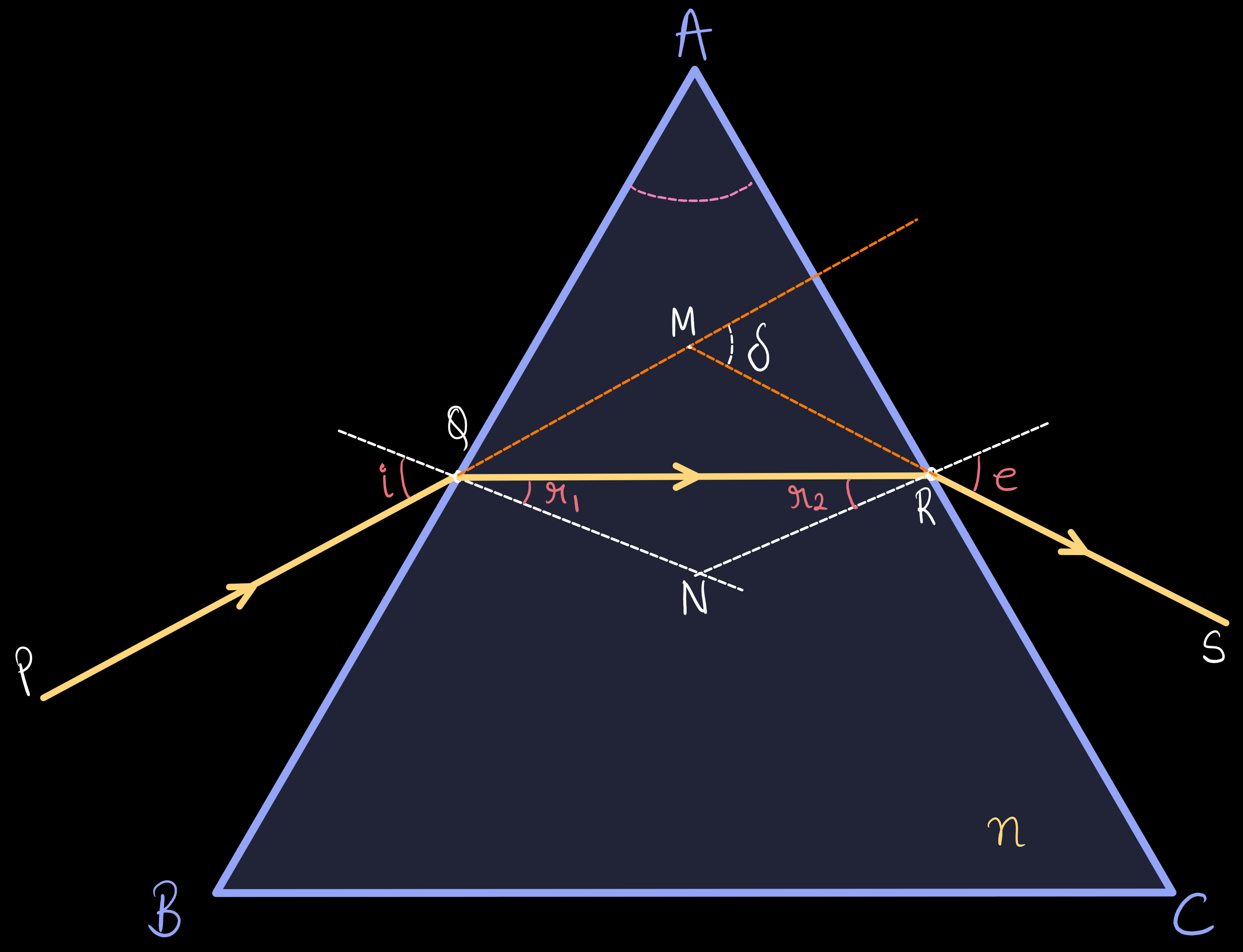
Refraction Through a Prism (Derivation)

Using Snell's at face AB -

$$1 \times \sin i = n \times \sin r$$

from eq^n (4) & (5)

$$i = \frac{A + \delta_m}{2} \quad \& \quad r = \frac{A}{2}$$



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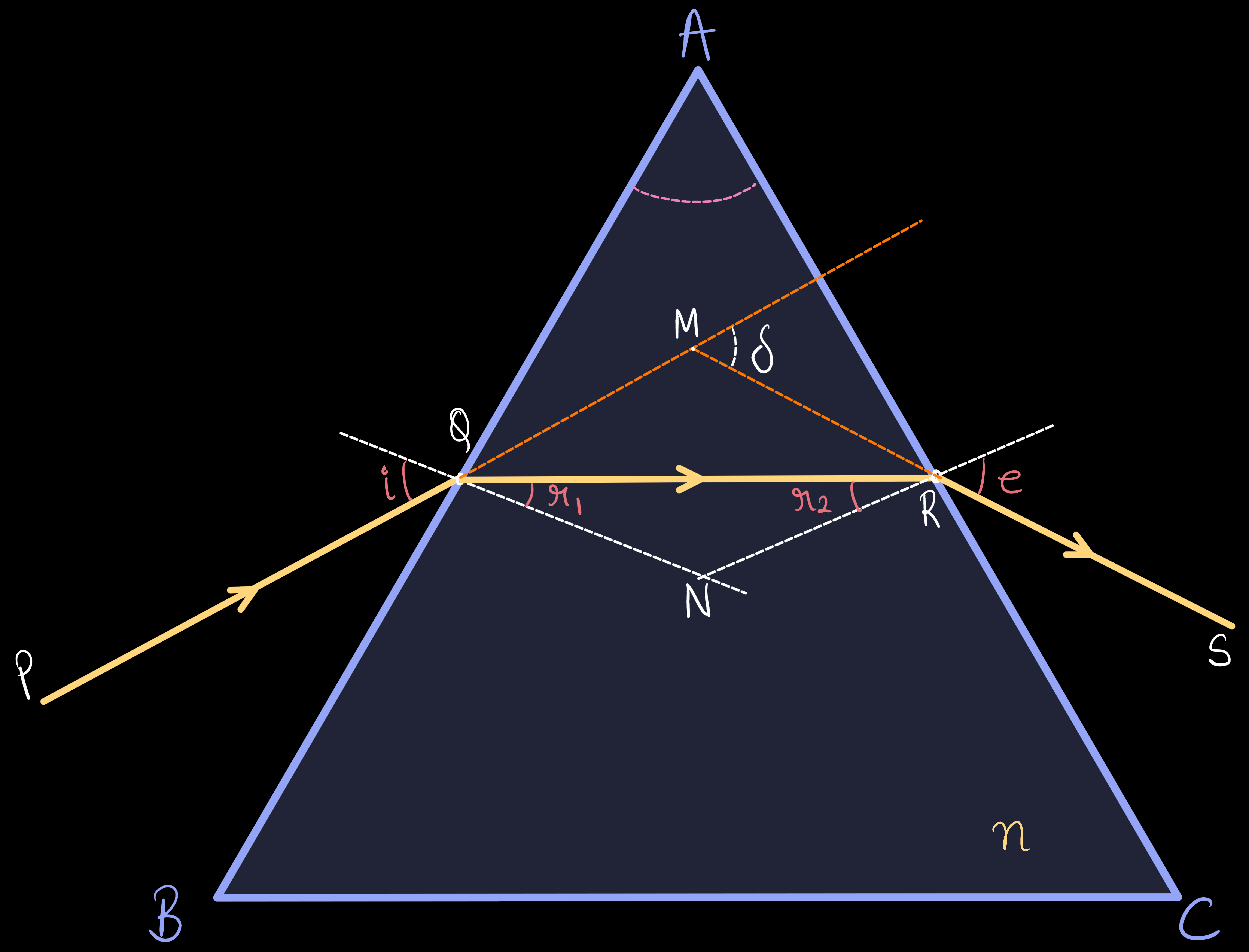
Using Snell's at face AB -

$$1 \times \sin i = n \times \sin r$$

from eq^n 4 & 5

$$i = \frac{A + \delta_m}{2} \quad \& \quad r = \frac{A}{2}$$

$$\therefore \quad \sin\left(\frac{A + \delta_m}{2}\right) = n \sin\left(\frac{A}{2}\right)$$



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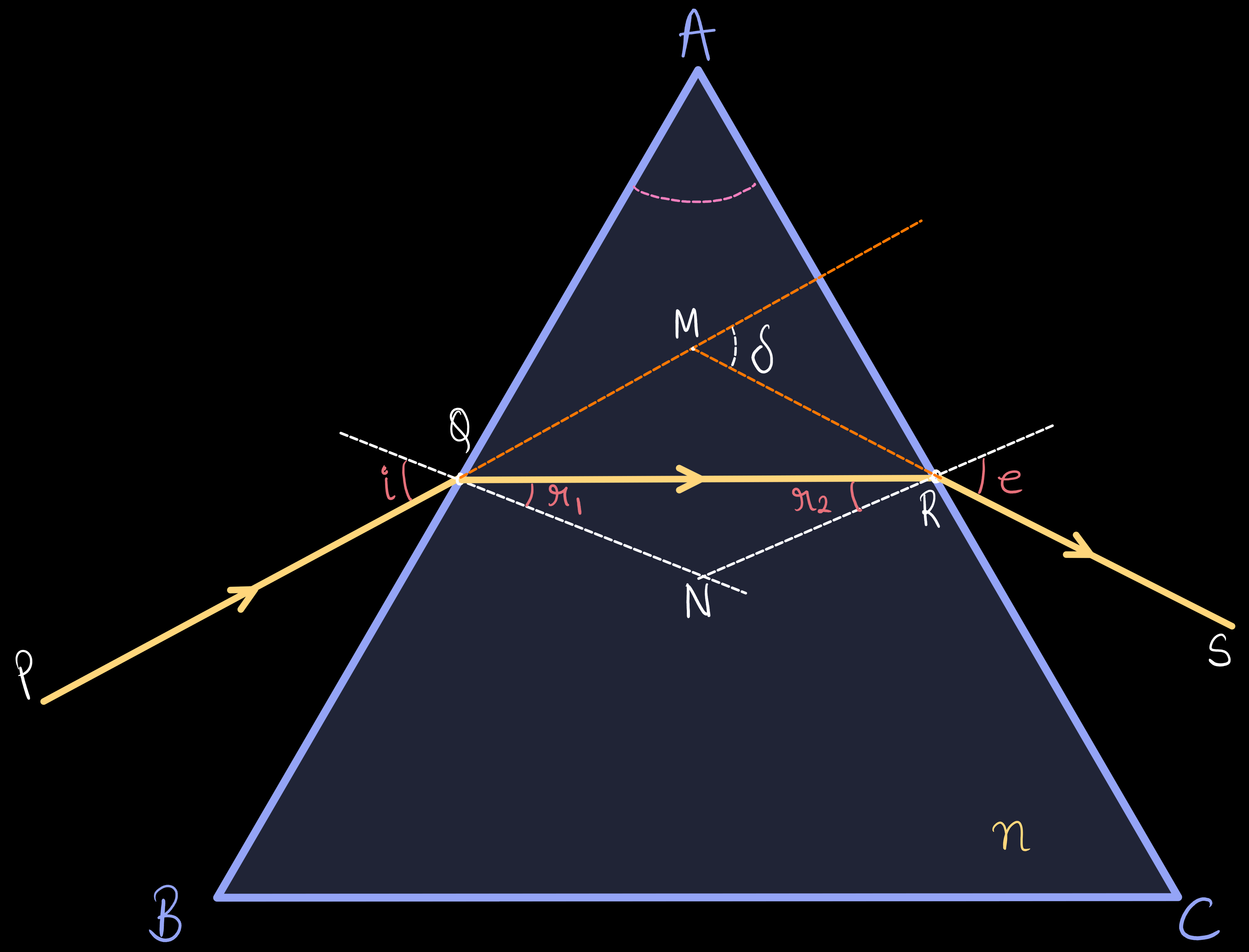

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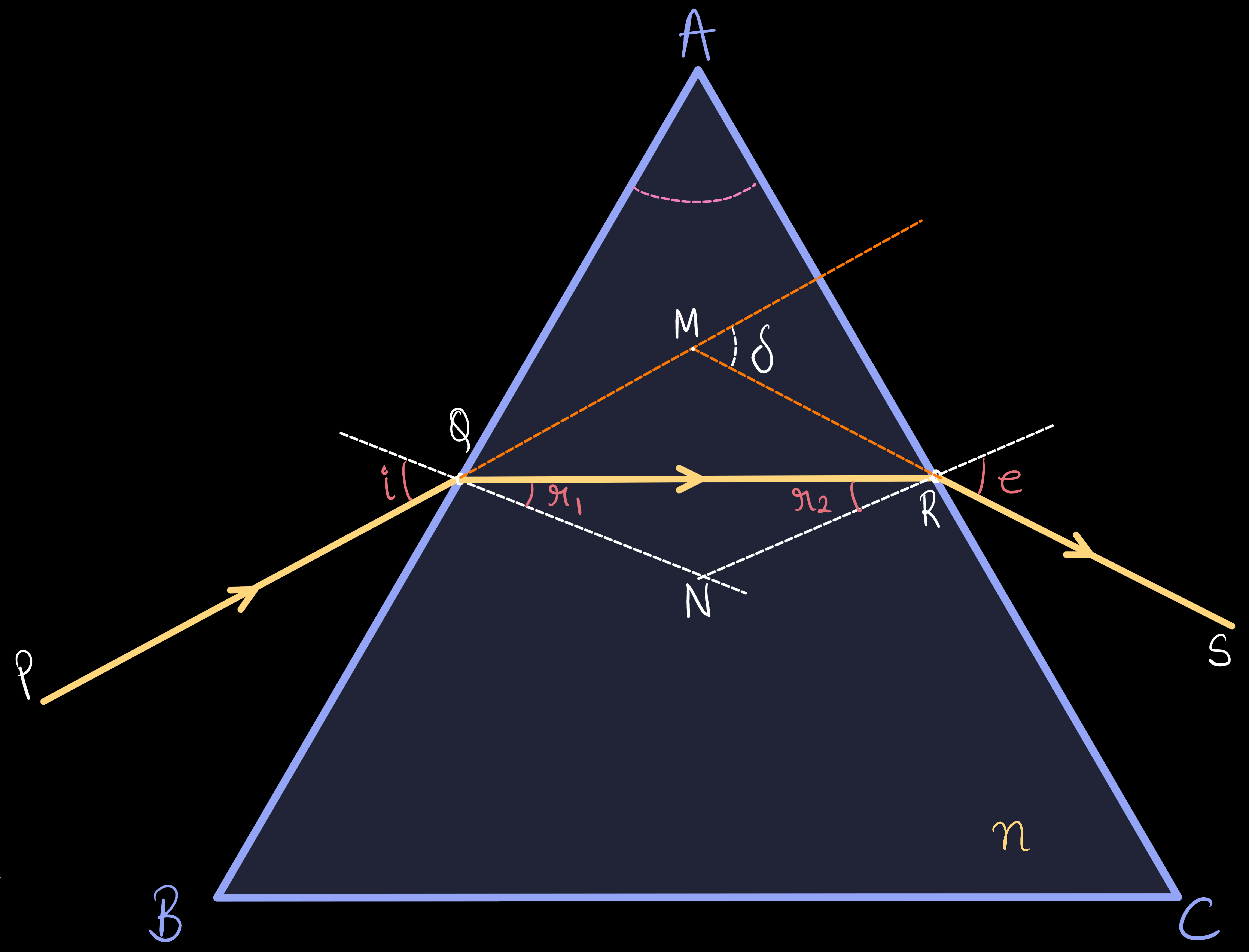
for small angle prism -

$$\sin\left(\frac{A+\delta_m}{2}\right) \approx \frac{A+\delta_m}{2}, \quad \sin\left(\frac{A}{2}\right) \approx \frac{A}{2}$$



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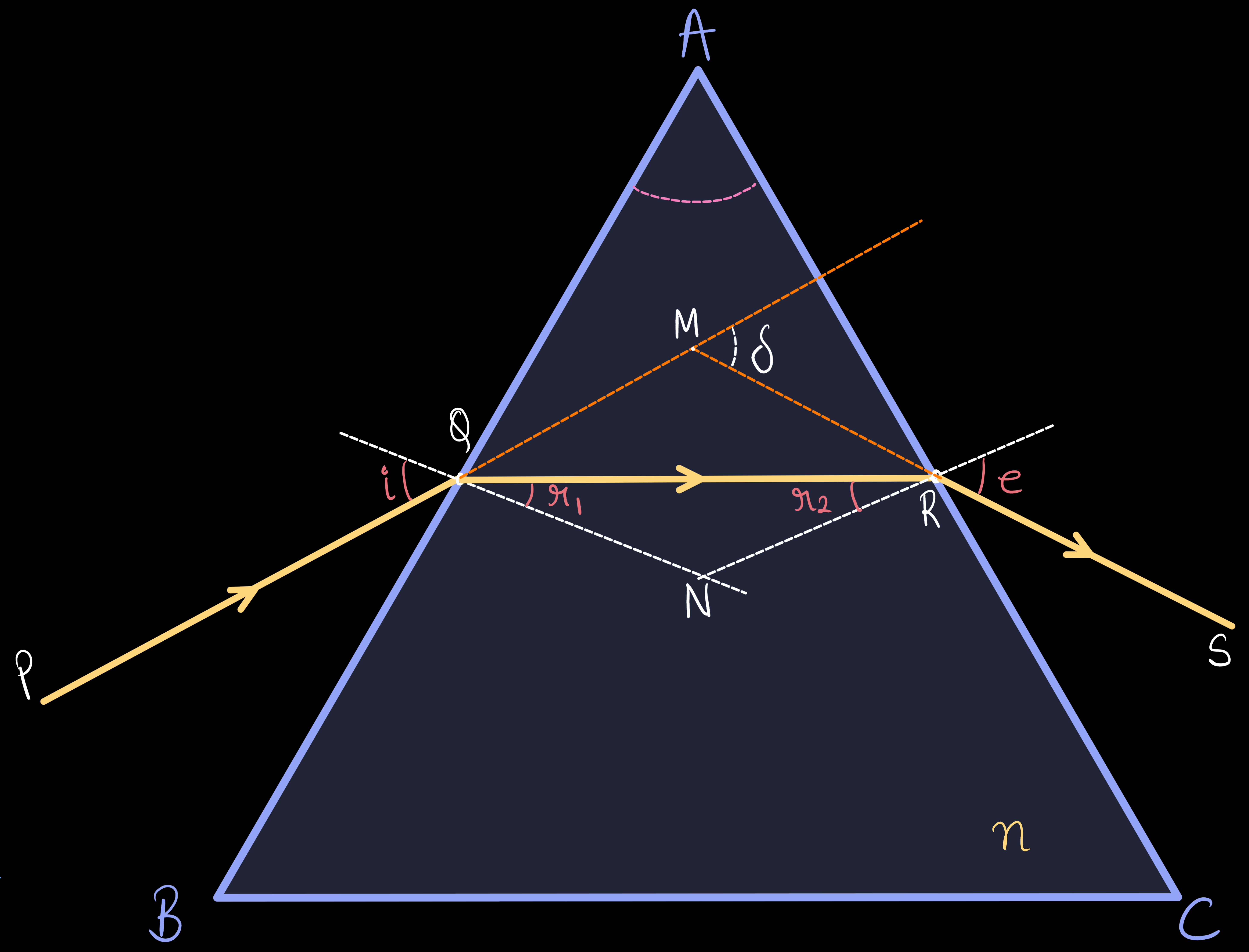

for small angle prism -

$$\sin\left(\frac{A+\delta_m}{2}\right) \approx \frac{A+\delta_m}{2}, \quad \sin\left(\frac{A}{2}\right) \approx \frac{A}{2}$$

$$n = \frac{A+\delta_m}{A} \Rightarrow \boxed{\delta_m = A(n-1)}$$